Revision – Sample Questions

1.

a. Form a linear program as follows

x – Product X
y – Product Y

maximise

$$24x + 23y - (3 \cdot 1 + 3 \cdot 4 + 4 \cdot 1)x - (2 \cdot 3 + 6 \cdot 1 + 4 \cdot 1)y = 5x + 7y$$

subject to:

$x + 3y \leq 60$ (Hours for A)
$4x + y \leq 75$ (Hours for B)
$x + y \leq 27$ (Hours for C)
and

$$x, y \geq 0$$

Solutions via Pyomo

```
Displaying solution ##################

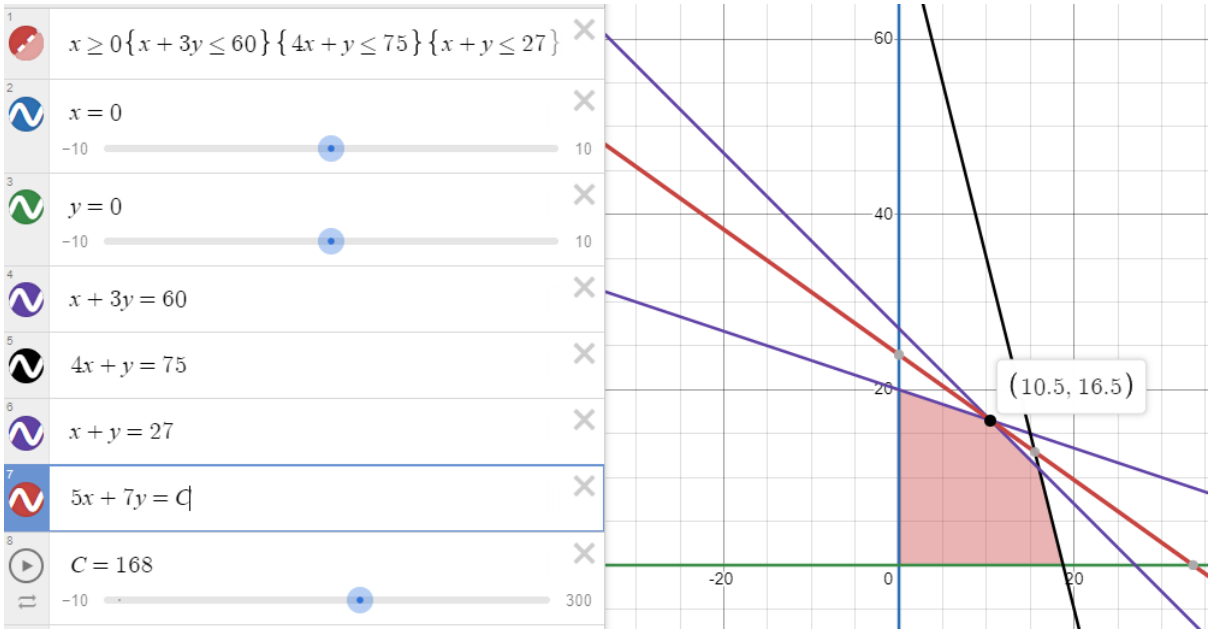
Model unknown

  Variables:
    x : Size=2, Index={0, 1}
        Key : Lower : Value : Upper : Fixed : Stale : Domain
          0 :     0 :  10.5 :  None : False : False : NonNegativeReals
          1 :     0 :  16.5 :  None : False : False : NonNegativeReals

  Objectives:
    obj : Size=1, Index=None, Active=True
        Key  : Active : Value
        None :   True : 168.0

  Constraints:
    constraint1 : Size=1
        Key  : Lower : Body : Upper
        None :  None : 60.0 :  60.0
    constraint2 : Size=1
        Key  : Lower : Body : Upper
        None :  None : 58.5 :  75.0
    constraint3 : Size=1
        Key  : Lower : Body : Upper
        None :  None : 27.0 :  27.0
```

Solution via Desmos



b. Add the additional constraint

$$x \geq \frac{3}{2}y$$

e.g. if $y = 4$ then $x \geq 6$. Hence the percentage of produce x is $\frac{6}{10} * 100 = 60\%$

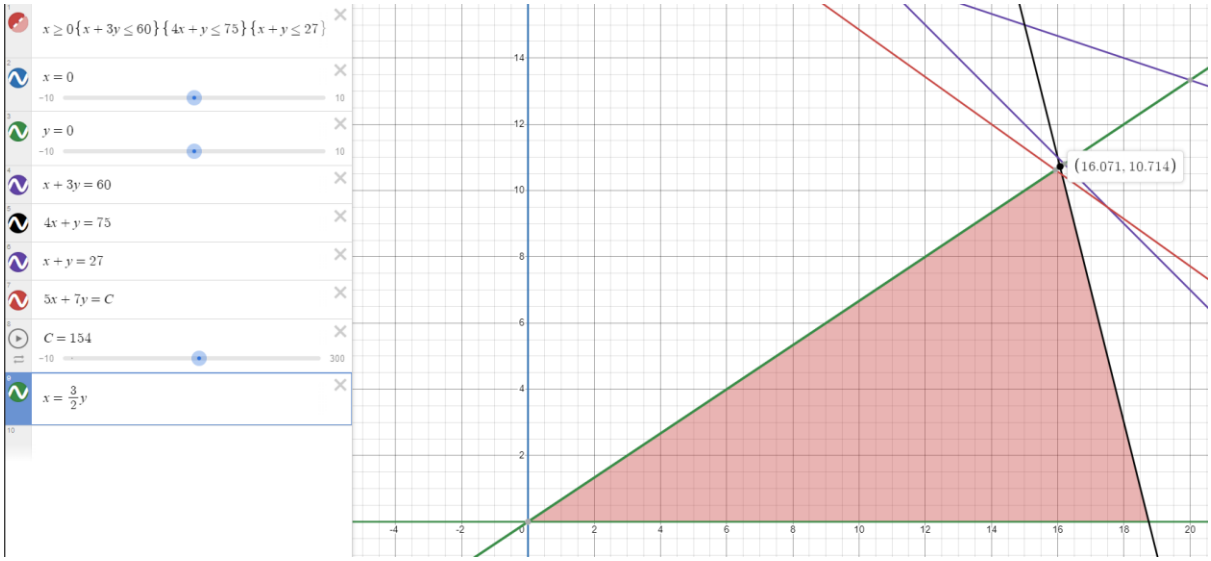
```
Displaying solution ##################

Model unknown

  Variables:
    x : Size=2, Index={0, 1}
        Key : Lower : Value            : Upper : Fixed : Stale : Domain
          0 :     0 : 16.0714285714286 :  None : False : False : NonNegativeReals
          1 :     0 : 10.7142857142857 :  None : False : False : NonNegativeReals

  Objectives:
    obj : Size=1, Index=None, Active=True
        Key  : Active : Value
        None :   True : 155.3571428571429

  Constraints:
    constraint1 : Size=1
        Key  : Lower : Body               : Upper
        None :  None : 48.214285714285694 :  60.0
    constraint2 : Size=1
        Key  : Lower : Body              : Upper
        None :  None : 75.0000000000001 :  75.0
    constraint3 : Size=1
        Key  : Lower : Body              : Upper
        None :  None : 26.7857142857143 :  27.0
    constraint4 : Size=1
        Key  : Lower : Body                    : Upper
        None :  None : -5.3290705182007514e-14 :   0.0
```



c. Add a constraint

$(3 \cdot 1 + 3 \cdot 4 + 4 \cdot 1)x + (2 \cdot 3 + 6 \cdot 1 + 4 \cdot 1)y \leq 400$ (Total hours cost for A, B and C)

```
Displaying solution ##################

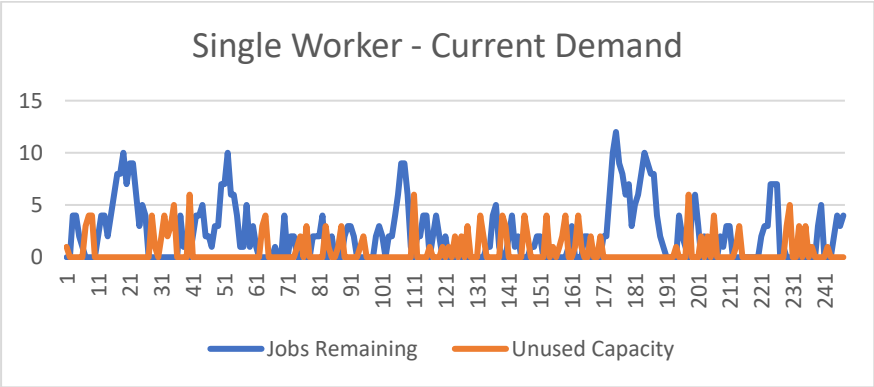
Model unknown

  Variables:
    x : Size=2, Index={0, 1}
        Key : Lower : Value             : Upper : Fixed : Stale : Domain
          0 :     0 : 13.4831460674157 :  None : False : False : NonNegativeReals
          1 :     0 : 8.98876404494382 :  None : False : False : NonNegativeReals

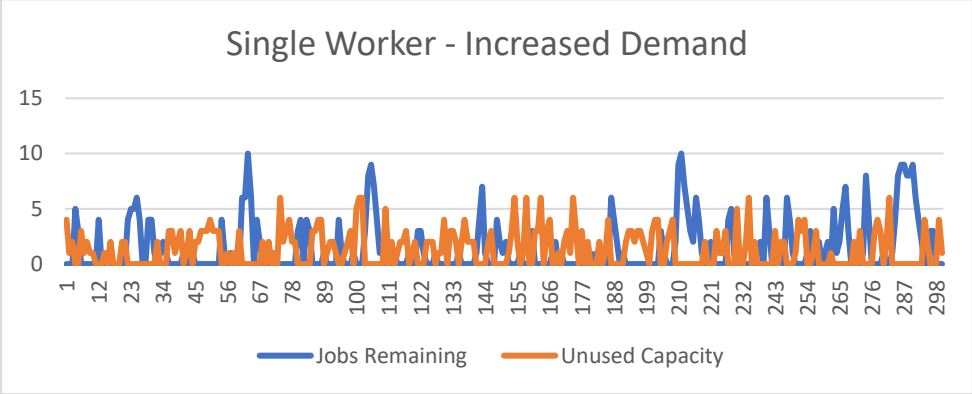
  Objectives:
    obj : Size=1, Index=None, Active=True
        Key  : Active : Value
        None :   True : 130.33707865168526

  Constraints:
    constraint1 : Size=1
        Key  : Lower : Body               : Upper
        None :  None : 40.449438202247165 :  60.0
    constraint2 : Size=1
        Key  : Lower : Body              : Upper
        None :  None : 62.92134831460662 :  75.0
    constraint3 : Size=1
        Key  : Lower : Body              : Upper
        None :  None : 22.47191011235952 :  27.0
    constraint4 : Size=1
        Key  : Lower : Body                   : Upper
        None :  None : 3.197442310920451e-14 :   0.0
    constraint5 : Size=1
        Key  : Lower : Body               : Upper
        None :  None : 399.99999999999943 : 400.0
```

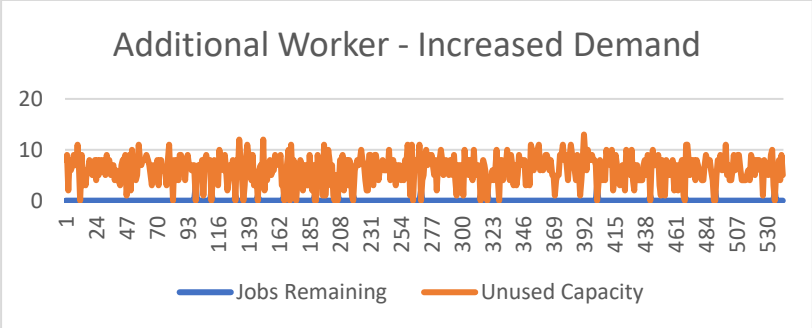
2.



This appears to be manageable no long term build up of jobs, some unused capacity, but clearly this does not affect the jobs being completed.

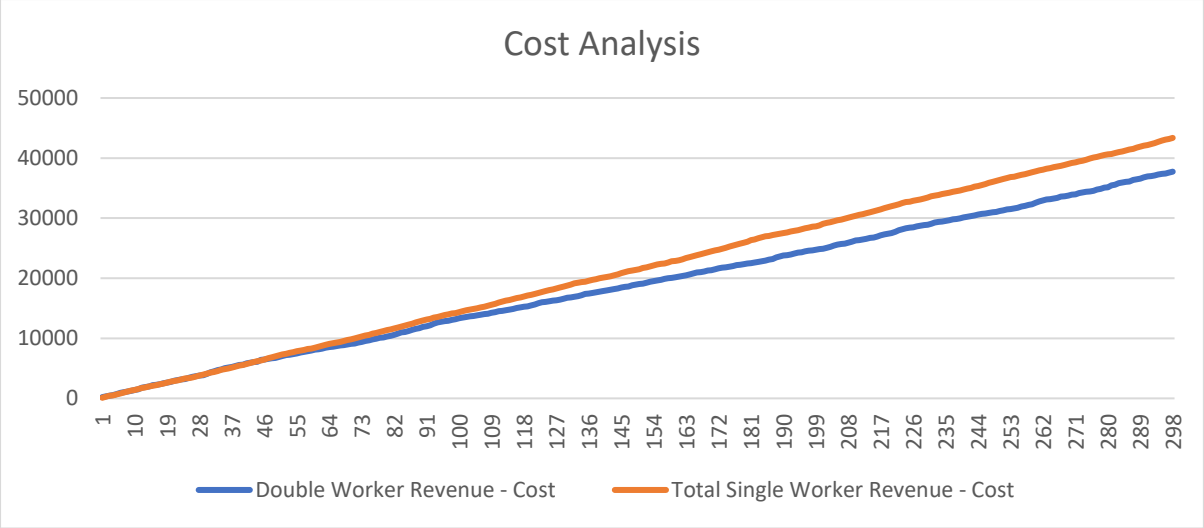


Again, this appears to be manageable no long term build up of jobs, some unused capacity, but clearly this does not affect the jobs being completed.



Whilst we would need to examine the long-term behaviour of the system, we can see the following:

- The job load for a single worker with increased demand appears manageable.
- In the case of the additional worker there appears to be a lot of unused capacity, i.e. idle time.



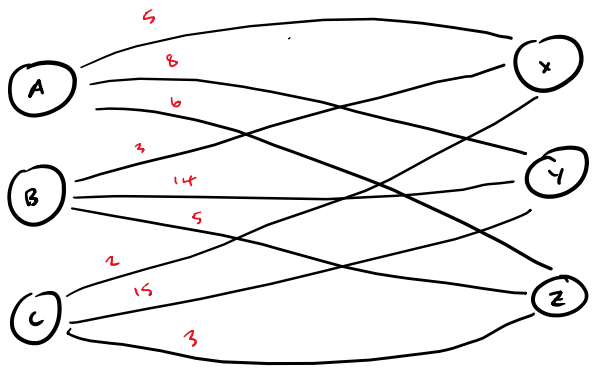
The cost analysis also suggests that a single worker is the best option.

If we analyse the expected increased demand and the expected workrate we see that the single worker has a value of 9.1 and the expected increased demand is 8.1. Hence a single worker is enough.

Interestingly the expected current demand of 8.4 is more than 8.1.

3.

a.



b.

```
Objective Value = 1200.0
Amount transported from SupplierA to ConsumerX: 0.0
Amount transported from SupplierA to ConsumerY: 80.0
Amount transported from SupplierA to ConsumerZ: 0.0
Amount transported from SupplierB to ConsumerX: 70.0
Amount transported from SupplierB to ConsumerY: 0.0
Amount transported from SupplierB to ConsumerZ: 10.0
Amount transported from SupplierC to ConsumerX: 0.0
Amount transported from SupplierC to ConsumerY: 0.0
Amount transported from SupplierC to ConsumerZ: 100.0
```

c.

```
Objective Value = 1230.0
Amount transported from SupplierA to ConsumerX: 10.0
Amount transported from SupplierA to ConsumerY: 80.0
Amount transported from SupplierA to ConsumerZ: 10.0
Amount transported from SupplierB to ConsumerX: 60.0
Amount transported from SupplierB to ConsumerY: 0.0
Amount transported from SupplierB to ConsumerZ: 0.0
Amount transported from SupplierC to ConsumerX: 0.0
Amount transported from SupplierC to ConsumerY: 0.0
Amount transported from SupplierC to ConsumerZ: 100.0
```

d.

Yes we should use as the objective is lower than previously

```
        ('A', 'D') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('A', 'X') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('A', 'Y') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('A', 'Z') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('B', 'D') :     0 : 100.0 :  None : False : False : NonNegativeReals
        ('B', 'X') :     0 :  60.0 :  None : False : False : NonNegativeReals
        ('B', 'Y') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('B', 'Z') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('C', 'D') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('C', 'X') :     0 :  10.0 :  None : False : False : NonNegativeReals
        ('C', 'Y') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('C', 'Z') :     0 :  90.0 :  None : False : False : NonNegativeReals
        ('D', 'X') :     0 :   0.0 :  None : False : False : NonNegativeReals
        ('D', 'Y') :     0 :  80.0 :  None : False : False : NonNegativeReals
        ('D', 'Z') :     0 :  20.0 :  None : False : False : NonNegativeReals

  Objectives:
    objective : Size=1, Index=None, Active=True
        Key  : Active : Value
        None :   True : 1050.0
```

4.

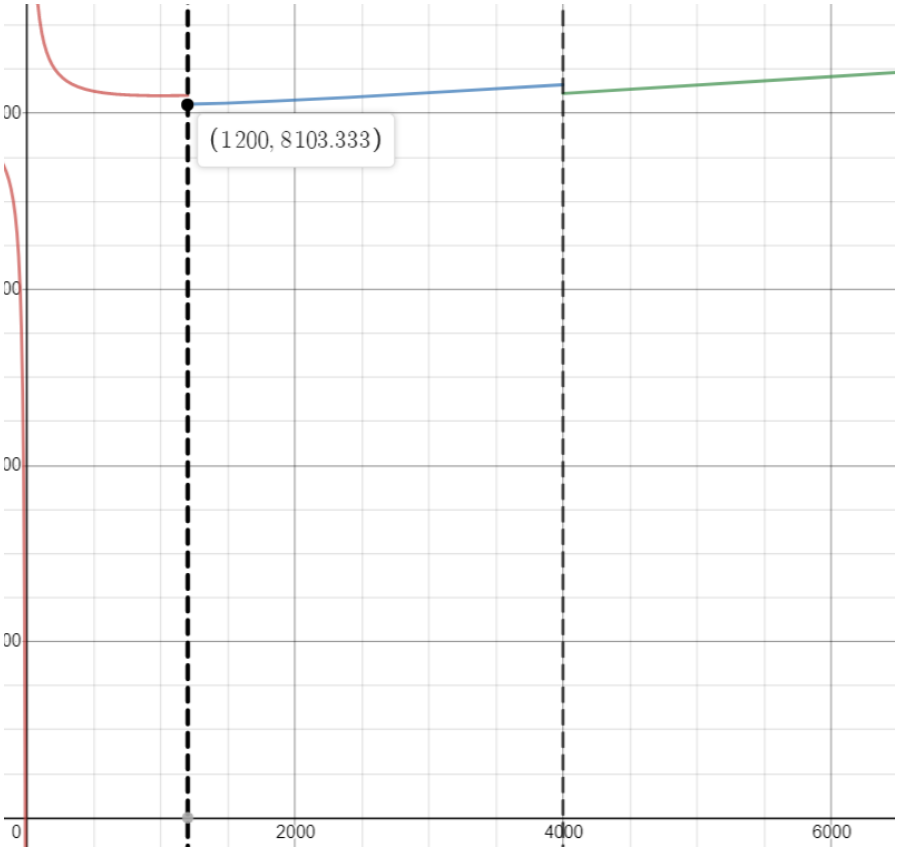
$$T(Q) = PD + \frac{KD}{Q} + \frac{hQ}{2}$$

$$EOQ = \frac{\sqrt{2KD}}{h} = \sqrt{\frac{2 \cdot 10 \cdot 10000}{0.2}} = 1000$$

As we have price breaks, the optimal order will occur at either EOQ or just past a price break larger than EOQ.

$$T(1000) = £8200$$
$$T(1200) = £8103.33$$
$$T(1400) = £8225$$

Thus we have a minimum, $Q^* = 1200, T(Q^*) = £8103.33$

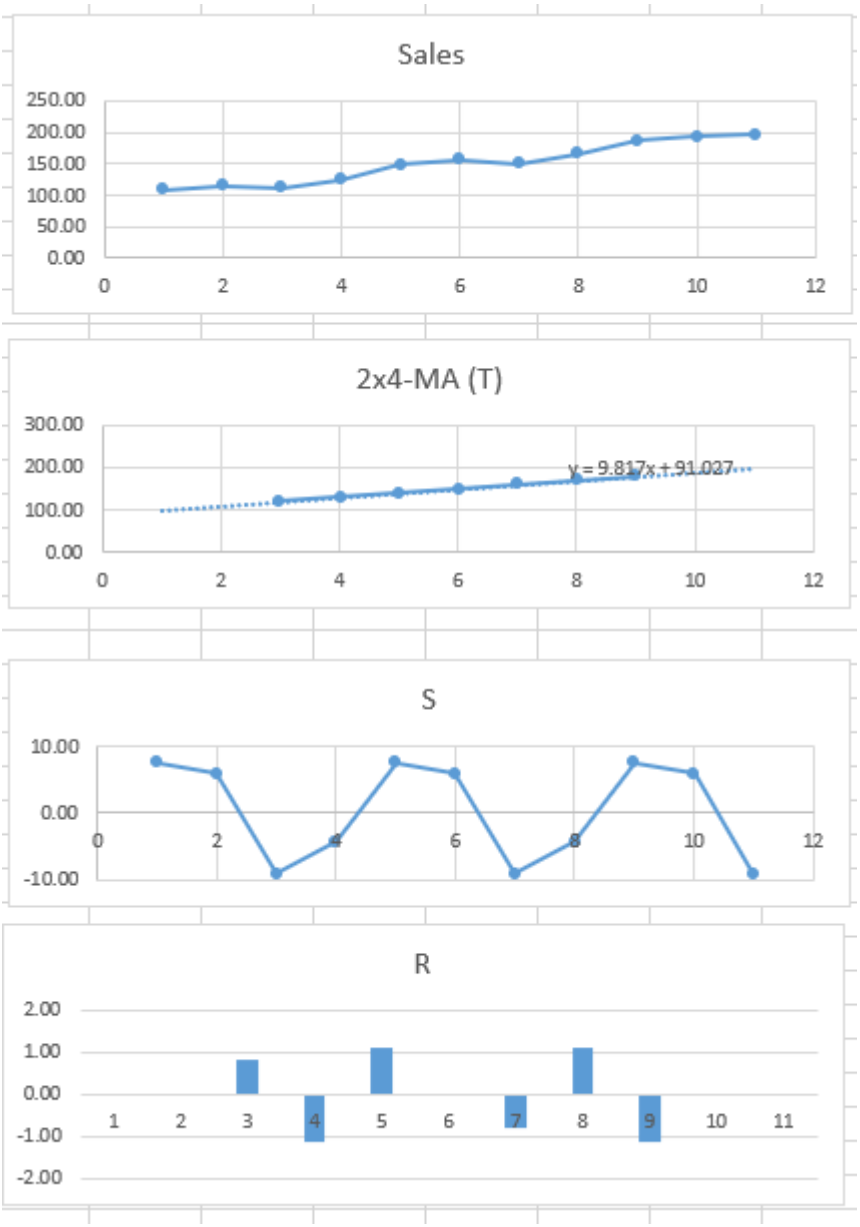


5.Assume Y=T+S+R

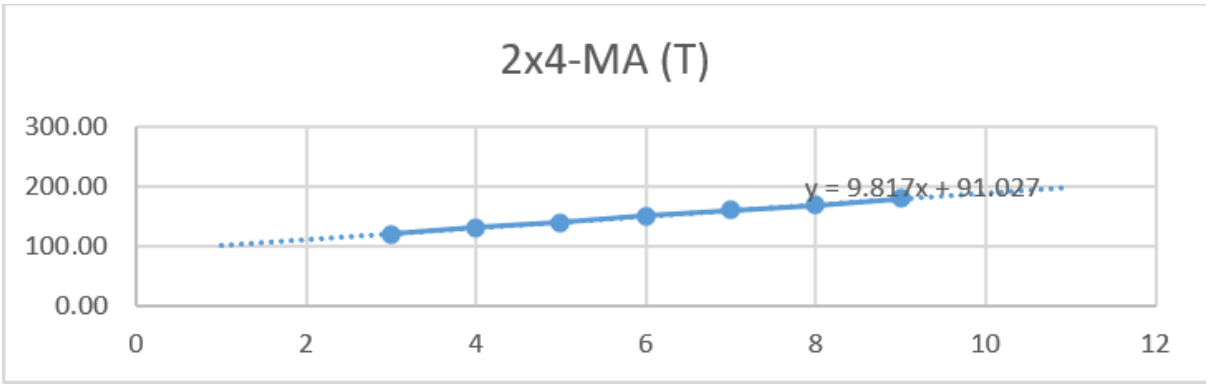
Compute a 2X4-MA to estimate the trend (T), then compute the seasonality (S) and remainder (R).

	Quarter	Sales (Y)	4-MA	2x4-MA (T)	Y-T	S	S Norm
1	1	109.00				7.63	7.58
2	2	115.00	115.25			5.88	5.83
3	3	112.00	125.25	120.25	-8.25	-9.06	-9.11
4	4	125.00	135.50	130.38	-5.38	-4.25	-4.30
5	1	149.00	145.00	140.25	8.75	7.63	7.58
6	2	156.00	155.25	150.13	5.88	5.88	5.83
7	3	150.00	164.50	159.88	-9.88	-9.06	-9.11
8	4	166.00	173.75	169.13	-3.13	-4.25	-4.30
9	1	186.00	185.25	179.50	6.50	7.63	7.58
10	2	193.00				5.88	5.83
11	3	196.00				-9.06	-9.11
12	4					-4.25	-4.30
13	1					7.625	7.58
14	2					5.875	5.83
15	3					-9.0625	-9.11

Thus, our additive decomposition looks as follows.



Fit a model to the trend, here we fit a linear model, if you are doing this by hand you would draw a straight line through the data.



The linear model for T is:

$$y = 9.817x + 91.027$$

Use this plus the seasonality to predict the future quarters…

	Quarter	Sales (Y)	4-MA	2x4-MA (T)	Y-T	S	S Norm	R	Linear	Linear + S
1	1	109.00				7.63	7.58		100.844	108.42
2	2	115.00	115.25			5.88	5.83		110.661	116.49
3	3	112.00	125.25	120.25	-8.25	-9.06	-9.11	0.86	120.478	111.37
4	4	125.00	135.50	130.38	-5.38	-4.25	-4.30	-1.08	130.295	126.00
5	1	149.00	145.00	140.25	8.75	7.63	7.58	1.17	140.112	147.69
6	2	156.00	155.25	150.13	5.88	5.88	5.83	0.05	149.929	155.76
7	3	150.00	164.50	159.88	-9.88	-9.06	-9.11	-0.77	159.746	150.64
8	4	166.00	173.75	169.13	-3.13	-4.25	-4.30	1.17	169.563	165.27
9	1	186.00	185.25	179.50	6.50	7.63	7.58	-1.08	179.38	186.96
10	2	193.00				5.88	5.83		189.197	195.03
11	3	196.00				-9.06	-9.11		199.014	189.90
12	4					-4.25	-4.30		208.831	204.53
13	1					7.625	7.58		218.648	226.23
14	2					5.875	5.83		228.465	234.29
15	3					-9.0625	-9.11		238.282	229.17

